Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 07)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 1 . 1 Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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0 1 . 2 Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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0 1 . 3 Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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0 1 . 4 Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

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0 2 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 2 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 2 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 3 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 3 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

0 3 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

0 3 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

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0 3 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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0 4 A new lowland nature reserve is being created on former farmland next to an existing ancient woodland.

0 4 . 1 Explain why the **area** and the **shape** of the reserve both matter in its design.

[3 marks]

0 4 . 2 Explain the value of a biological corridor linking the new reserve to the ancient woodland.

[2 marks]

0 4 . 3 Explain how each of the following abiotic factors could be managed to increase the biodiversity of the reserve.

[4 marks]

Water levels .....

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Soil nutrient levels .....

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Topography .....

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Light levels .....

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0 4 . 4 Much of the reserve will be kept as a **plagioclimax**. Explain what this means and how grazing is used to maintain one.

[3 marks]

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0 4 . 5 Explain why the removal or culling of some species may be necessary on the reserve.

[3 marks]

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0 5 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 5 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 5 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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0 6 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	**Biocapacity / gha per person**	**Human Development Index**
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

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0 6 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 6 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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0 6 . 3 Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer _____ Earths

0 6 . 4 Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 6 . 5 Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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0 7 . 1 Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

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0 7 . 2 Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

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0 7 . 3 Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

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0 7 . 4 Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

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0 8 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 8 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 9 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

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0 9 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

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0 9 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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1 0 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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1 0 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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1 0 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

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1 0 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS